Pen and Paper Task 1 (5 points)

Given the following documents:

• Document 1: water nice

• Document 2: water

• Document 3: water

• Document 4: water

• Document 5: sun nice

• Document 6: sun is nice

• Document 7: water sun

• Document 8: sun

• Document 9: sun

Subtask 1.1

Build the full inverted index (without positional information) for these documents. Do not perform normalization of terms.

Answer:

Steps to build inverted index:

1. Tokenization and Preprocessing : Skipped (no normalization)
2. Generating postings :

Term	Document ID
water	1
nice	1
water	2
water	3
water	4
sun	5
nice	5
sun	6
is	6
nice	6
water	7
sun	7
sun	8
sun	9

3. Sort Postings alphabetically

Term	Document ID
is	6
nice	1
nice	5
nice	6
sun	5
sun	6
sun	7
sun	8
sun	9
water	1
water	2
water	3
water	4
water	7

4. Create Postings list

Term	Frequency	Postings List
is	1	[6]
nice	3	[1] -> [5] -> [6]
sun	5	[5] -> [6] -> [7] -> [8] -> [9]
water	5	[1] -> [2] -> [3] -> [4] -> [7]

Subtask 1.2

Add skip pointers to the index from the previous task. Provide an example query which demonstrates the usefulness of skip pointers for your index and explain wh